

Project Overview

HandsMen Threads is a Salesforce lightning app project that will make it easier for the team to manage information and create a smoother experience for customers. The project centers on building a dependable data system that keeps everything accurate from the moment it's entered, helping the business avoid confusion and make clearer decisions. It also brings in helpful automations like sending customers order confirmation emails, updating loyalty status based on their purchases, alerting the warehouse when stock is running low, and processing bulk orders each night to keep inventory and finances up to date. Altogether, this project is designed to make daily work simpler, improve communication, and support better service across the company.

Objectives

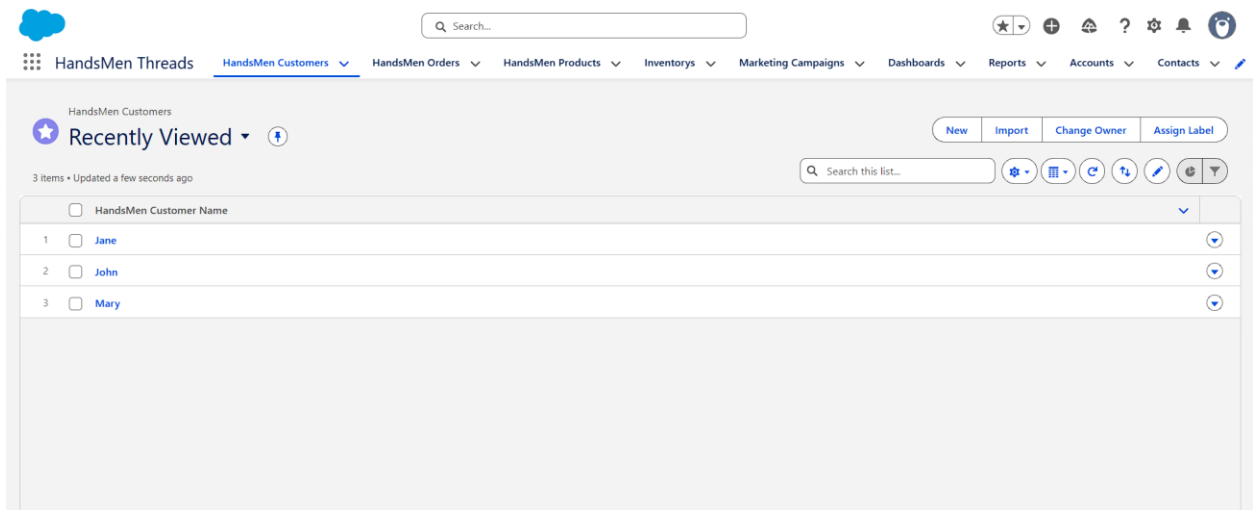
- **Centralized Data Management:** To consolidate Customer, Product, and Service Provider (Handsmen) data into a single, accessible source of truth.
- **Automated Logic:** To remove human error in financial calculations and stock adjustments using Formula Fields and Apex triggers.
- **Customer Retention:** To implement an automatic "Loyalty Tier" system (Gold/Silver/Bronze) that rewards high-value clients without manual intervention.

Phase 1: Requirement Analysis & Planning

The primary challenge addressed by this project is the inefficiency of manual inventory tracking and order processing. The business required a system that could automatically alert staff when stock runs low and ensure that "Total Amount" calculations are always mathematically correct based on quantity and price.

To support these requirements, a custom Schema was designed consisting of five core objects:

1. **Handsmen Customer:** Serves as the master record for client data and loyalty status tracking.
2. **Handsmen Product:** Functions as the catalog for goods and services, including pricing and SKU details.
3. **Inventorys:** A dedicated object for tracking real-time stock levels and warehouse data.
4. **Handsmen Order:** The transactional junction object that links Customers, Products, and Handsmen to process a sale.
5. **Marketing Campaign:** Manages outreach efforts to drive sales.



Phase 2: Salesforce Development - Backend & Configurations

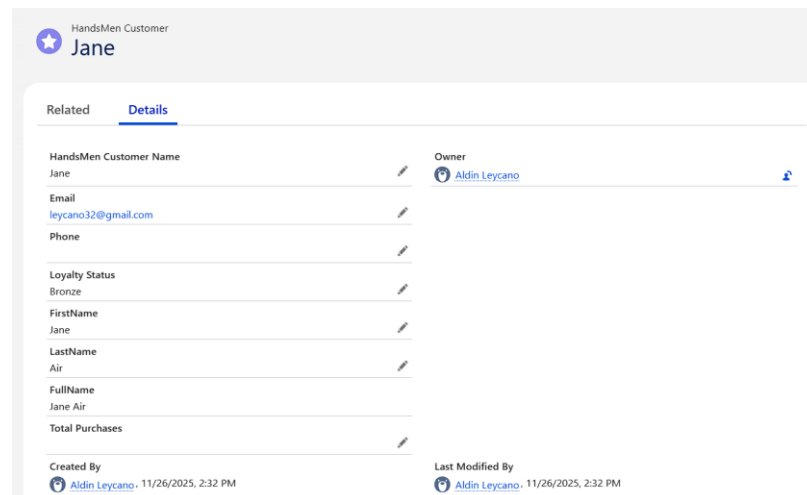
Customization of Objects & Fields

The data model was built to ensure data quality at the point of entry.

1. Formula Fields (Automation)

To reduce manual data entry, the system uses formulas to concatenate names and calculate costs automatically.

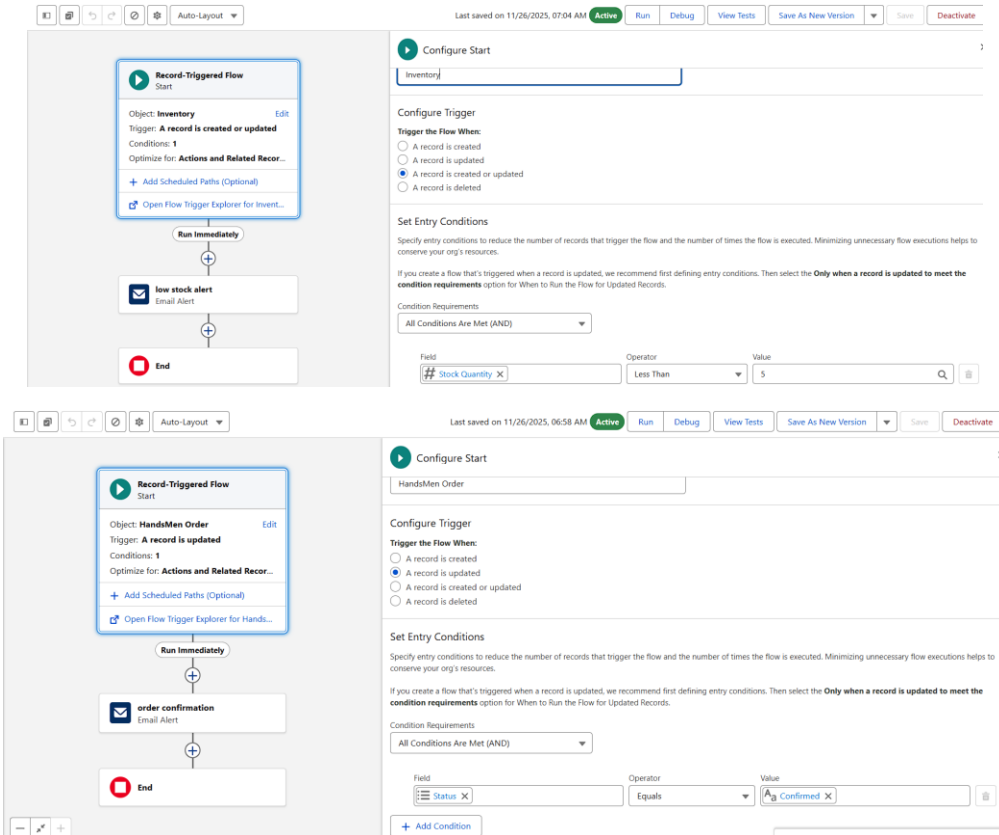
- *Identity Logic:* Full Name automatically combines First Name & " " & Last Name.
- *Financial Logic:* Total Amount is automatically calculated as Quantity * Price.



Automation Implementation (Flows)

1. Communication Automations

- **Low Stock Alert:** A Record-Triggered Flow monitors the Inventory's object. If the stock count drops below the defined threshold, an email alert is immediately sent to the warehouse manager.
- **Order Confirmation:** Upon the creation of a new Handsmen Order, a flow triggers an email to the customer confirming their purchase details.



2. Loyalty Program Logic

A sophisticated logic flow assigns a loyalty status based on the Total Purchases field on the Customer object:

- **Gold Status:** Triggered when purchases are above 1000.
- **Silver Status:** Triggered when purchases are between 500 and 1000.
- **Bronze Status:** Assigned for purchases between 0 and 499.

OUTCOME ORDER

Gold

Bronze

Silver

OUTCOME DETAILS

Delete Outcome

* Outcome Label

Gold

* Outcome API Name

Gold

Condition Requirements to Execute Outcome

All Conditions Are Met (AND)

* Resource

...ugh records > Total Purchases

Operator

Greater Than

Value

1000

OUTCOME ORDER

Gold

Bronze

Silver

OUTCOME DETAILS

Delete Outcome

* Outcome Label

Bronze

* Outcome API Name

Bronze

Condition Requirements to Execute Outcome

All Conditions Are Met (AND)

* Resource

...ugh records > Total Purchases

Operator

Less Than or Equal

Value

500

OUTCOME ORDER

Gold

Bronze

Silver

OUTCOME DETAILS

* Label

Silver



Programmatic Automation (Apex)

To handle complex transaction logic, an Apex Trigger was developed for Inventory Management.

```
OrderTotalTrigger.apex | StockDeductionTrigger.apex
Code Coverage: None | API Version: 65
1 trigger OrderTotalTrigger on HandsMen_Order__c (before insert, before update) {
2     Set<Id> productIds = new Set<Id>();
3
4     for (HandsMen_Order__c order : Trigger.new) {
5         if (order.HandsMen_Product__c != null) {
6             productIds.add(order.HandsMen_Product__c);
7         }
8     }
9
10    Map<Id, HandsMen_Product__c> productMap = new Map<Id, HandsMen_Product__c>{
11        [SELECT Id, Price__c FROM HandsMen_Product__c WHERE Id IN :productIds]
12    };
13
14    for (HandsMen_Order__c order : Trigger.new) {
15        if (order.HandsMen_Product__c != null && productMap.containsKey(order.HandsMen_Product__c)) {
16            HandsMen_Product__c product = productMap.get(order.HandsMen_Product__c);
17            if (order.Quantity__c != null) {
18                order.Total_Amount__c = order.Quantity__c * product.Price__c;
19            }
20        }
21    }
22 }
```

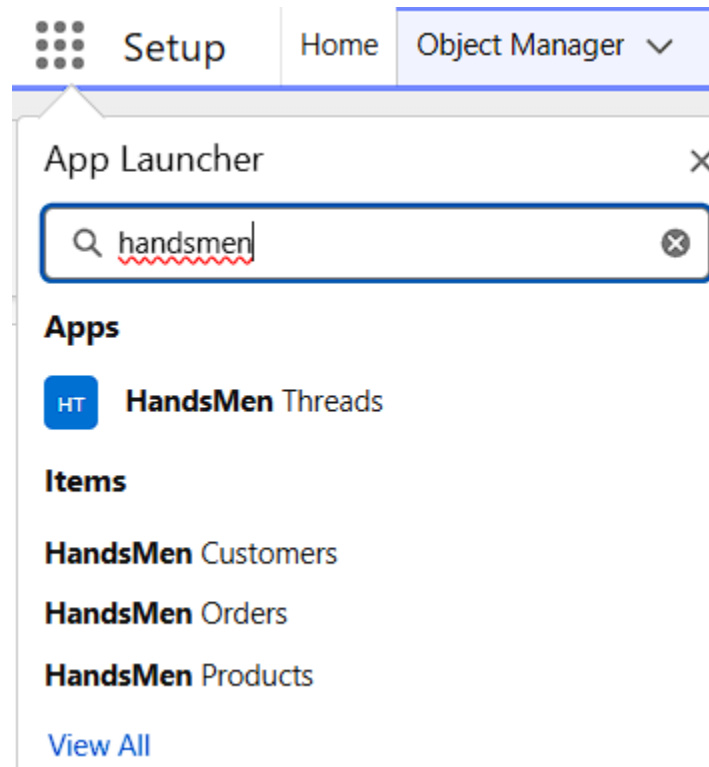
- Logic: When a Handsmen Order is finalized, the Apex code queries the related Inventories record and subtracts the order Quantity from the Current Stock. This ensures the database reflects physical reality in real-time.

```
OrderTotalTrigger.apex | StockDeductionTrigger.apex
Code Coverage: None | API Version: 65
1 trigger StockDeductionTrigger on HandsMen_Order__c (after insert, after update) {
2     Set<Id> productIds = new Set<Id>();
3
4     for (HandsMen_Order__c order : Trigger.new) {
5         if (order.Status__c == 'Confirmed' && order.HandsMen_Product__c != null) {
6             productIds.add(order.HandsMen_Product__c);
7         }
8     }
9
10    if (productIds.isEmpty()) return;
11
12    // Query related inventories based on product
13    Map<Id, Inventory__c> inventoryMap = new Map<Id, Inventory__c>{
14        [SELECT Id, stock_Quantity__c, HandsMen_Product__c
15         FROM Inventory__c
16         WHERE HandsMen_Product__c IN :productIds]
17    };
18
19    List<Inventory__c> inventoriesToUpdate = new List<Inventory__c>();
20
21    for (HandsMen_Order__c order : Trigger.new) {
22        if (order.Status__c == 'Confirmed' && order.HandsMen_Product__c != null) {
23            for (Inventory__c inv : inventoryMap.values()) {
24                if (inv.HandsMen_Product__c == order.HandsMen_Product__c) {
25                    inv.Stock_Quantity__c -= order.Quantity__c;
26                    inventoriesToUpdate.add(inv);
27                    break;
28                }
29            }
30        }
31    }
32
33    if (!inventoriesToUpdate.isEmpty()) {
34        update inventoriesToUpdate;
35    }
36 }
```

Phase 3: UI/UX Development & Customization

Lightning App Configuration

A custom "Handsmen App" was created in the Lightning App Manager. This combines all custom tabs (Customers, Orders, Inventories) into a single navigation bar, allowing users to switch contexts without leaving the application.



User-Friendly Page Layouts - The Record Pages were customized using the Lightning App Builder to prioritize essential information. The "Details" tab was reorganized to place the "Total Amount" and "Loyalty Status" fields at the top of the page for immediate visibility.

Related	Details
HandsMen Customer Name	John
Email	leycano32@gmail.com
Phone	(234) 567-8910
Loyalty Status	Bronze
FirstName	John
LastName	Doe
FullName	John Doe
Total Purchases	500
Created By	Aldin Leycano
	11/24/2025, 5:49 PM

Phase 4: Data Migration, Testing & Security

Security Model

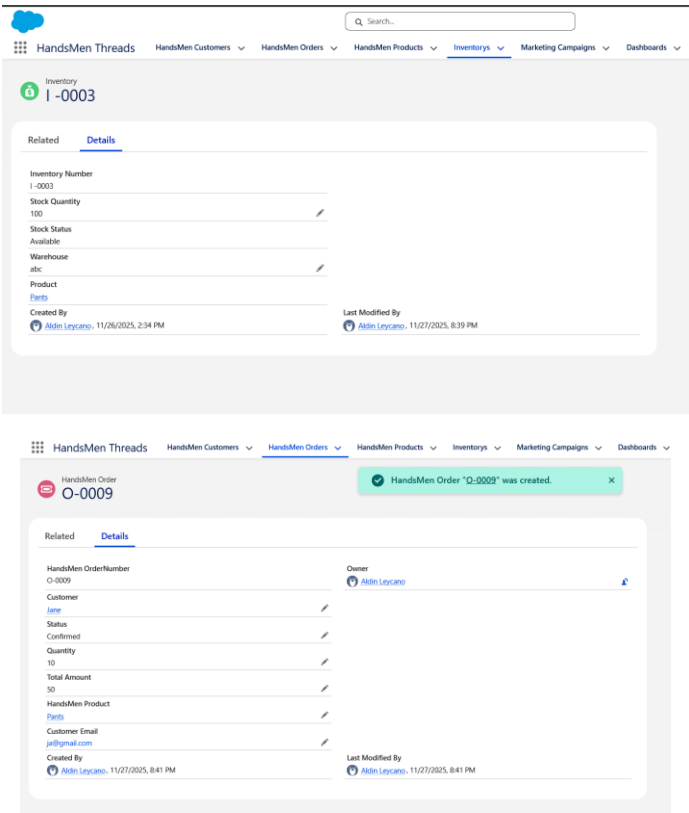
Access is controlled using a combination of Profiles and Permission Sets.

- Profiles: Standard users have "Read/Create" access to Orders.
- Permission Sets: Only specific managers are granted the "Delete" permission on Inventory records to prevent accidental data loss.

Testing & Validation (Test Cases)

Test Case 1: Apex Inventory Automation

- Objective: Verify stock is deducted upon order.
 - Input: Product "pants" has 100 stocks. User orders 10 pants.
- Output: Product "pants" inventory automatically updates to 90.



Related Details

Inventory Number

I -0003

Stock Quantity

90

Stock Status

Available

Warehouse

abc

Product

[Pants](#)

Created By

 [Aldin Leycano](#), 11/26/2025, 2:34 PM

Last Modified By

 [Aldin Leycano](#), 11/27/2025, 8:41 PM

Dear Jane,

Your order #O-0007 has been confirmed!

Thank you for shopping with us.

Best Regards,

Sales Team

Test Case 2: Loyalty Automation

- Objective: Verify Gold Tier assignment.
- Input: User creates a purchase of 1200.
- Output: Customer status changes to "Gold."



Congratulations! You are now a Gold member and you are eligible for our Loyalty Rewards Program.
Enjoy exclusive discounts, early access to offers, and special member benefits.
Thank you for your continued Support.



HandsMen Threads

HandsMen Customers

HandsMen Orders

HandsMen Products

Inventories

Marketing Campaigns

Dashboards

HandsMen Customer

Mary

Related

Details

HandsMen Customer Name

Mary

Email

mp@gmail.com

Phone

(234) 567-8910

Loyalty Status

Gold

FirstName

Mary

LastName

Pops

FullName

Mary Pops

Total Purchases

1,200

Created By

Aldin Leycano, 11/25/2025, 3:51 PM

Owner

Aldin Leycano

Last Modified By

Aldin Leycano, 11/27/2025, 8:43 PM

Phase 5: Deployment, Documentation & Maintenance

Deployment Strategy To safely move the Handsmen application from the building stage to the live business environment, we utilize a method called Change Sets. This process allows us to bundle all the specific components created such as the Handsmen Customer object, the Inventories object, and the Apex code into a single package. We then upload this package from the Sandbox environment to the Production organization. Before the final move, we always run a validation check. This step is crucial because it runs all our tests to ensure the code coverage is high enough and that no validation rules will break the existing data. Once the validation passes, we deploy the app, and then manually handle final touches like uploading the initial product list or assigning user permissions.

System Maintenance & Monitoring Keeping the system running smoothly requires regular attention even after it is launched. The main part of maintenance involves monitoring the automation to ensure it triggers correctly. For example, we regularly check the "Apex Jobs" list in the setup menu to confirm that the code responsible for deducting stock quantity runs successfully every time an order is confirmed. We also monitor the email logs to make sure that the Low Stock Alerts and Order Confirmation emails are actually being delivered to the users' inboxes and not getting stuck. Periodically, we also perform data cleaning to remove any duplicate customer records or outdated marketing campaigns, which helps keep the system fast and easy to use.

Troubleshooting Documentation If a user reports an issue, such as an error when trying to save a new order, we follow a strict troubleshooting process. First, we look at the specific error message displayed on the screen or review the Debug Logs to understand what went wrong. We never try to fix these issues directly in the live system to avoid causing more problems. Instead, we reproduce the error in our Sandbox environment to test potential fixes. Once we find a solution perhaps by adjusting a Validation Rule that was too strict, we save the change and use a Change Set to deploy the fix back to the live system.

Conclusion

The Handsmen Threads project has successfully modernized the way we manage our service and inventory operations. Before this application, tracking customers, orders, and stock levels was likely a manual and error-prone process. By centralizing everything into the Salesforce cloud, we have achieved our primary objectives. We now have a system where data integrity is guaranteed, for example, the validation rules ensure no one can enter a negative stock number or an invalid email address.

Furthermore, the automation we built has significantly reduced manual work. The system now automatically calculates the "Total Amount" of an order and instantly subtracts the sold items from the "Inventories" object, ensuring our stock counts are always accurate in real-time. The addition of the automatic Loyalty Program is a major improvement, as it instantly rewards our best customers with Gold, Silver, or Bronze status based on their purchase history without any staff

needing to do the math. Overall, the Handsmen app has made our business operations faster, more accurate, and much easier to manage.

Recommendations & Future Enhancements

To make the Handsmen system even more powerful in the future, there are several features we can recommend adding. First, integrating the Salesforce Mobile App would be a great step. This would allow Handsmen service providers to update job statuses or check inventory levels right from their phones while they are out working, rather than waiting to get back to a computer.

Additionally, we could implement Salesforce Einstein (AI) to help with business planning. Einstein could analyze our "Marketing Campaign" and "Order" data to predict which products will be in high demand next season, allowing us to order stock before we run out. Finally, creating a Web-to-Lead form on a public website would allow new customers to inquire about services online, and their information would automatically appear in Salesforce, creating a seamless bridge between our marketing and our database.